# Introduction to RStudio

## Day 1 - Introduction to Data Analysis with R

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# Difference between R and RStudio

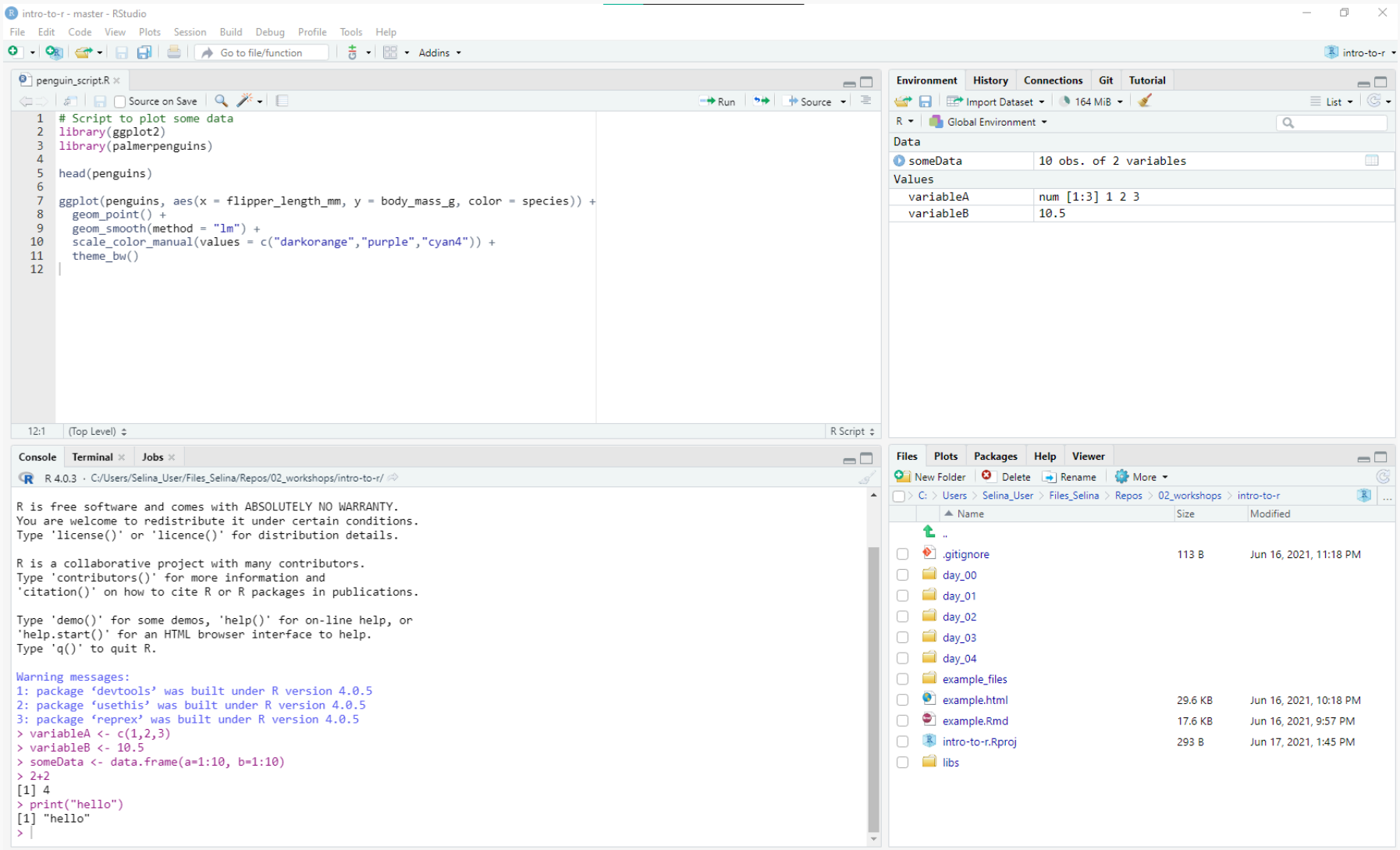
Analogy and image from ModernDive Book

R is the **programming language** and the **program** that does the actual work

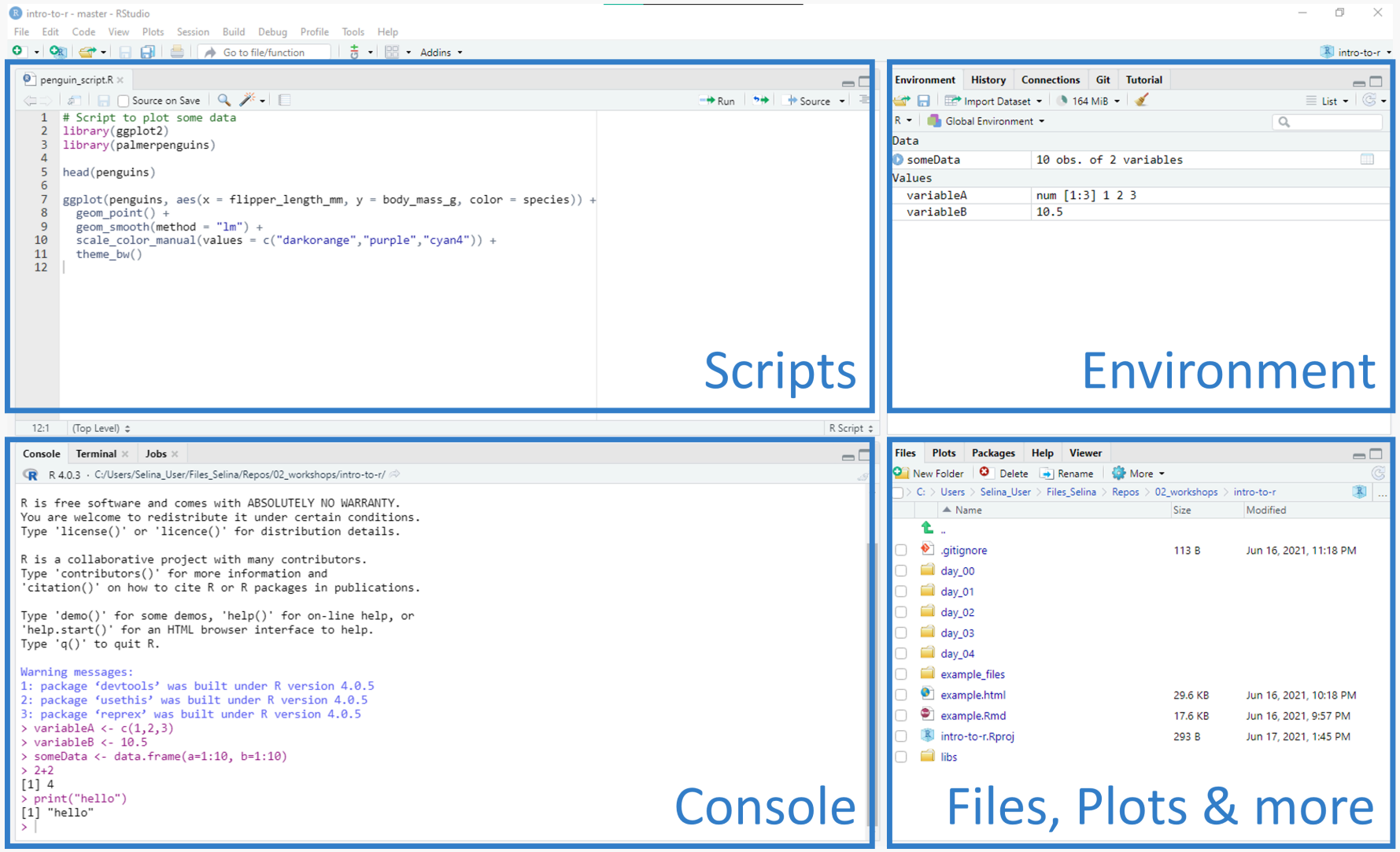
RStudio is the **integrated development environment** (IDE).

Interface to R, syntax highlighting, file management, …

# A quick tour around RStudio



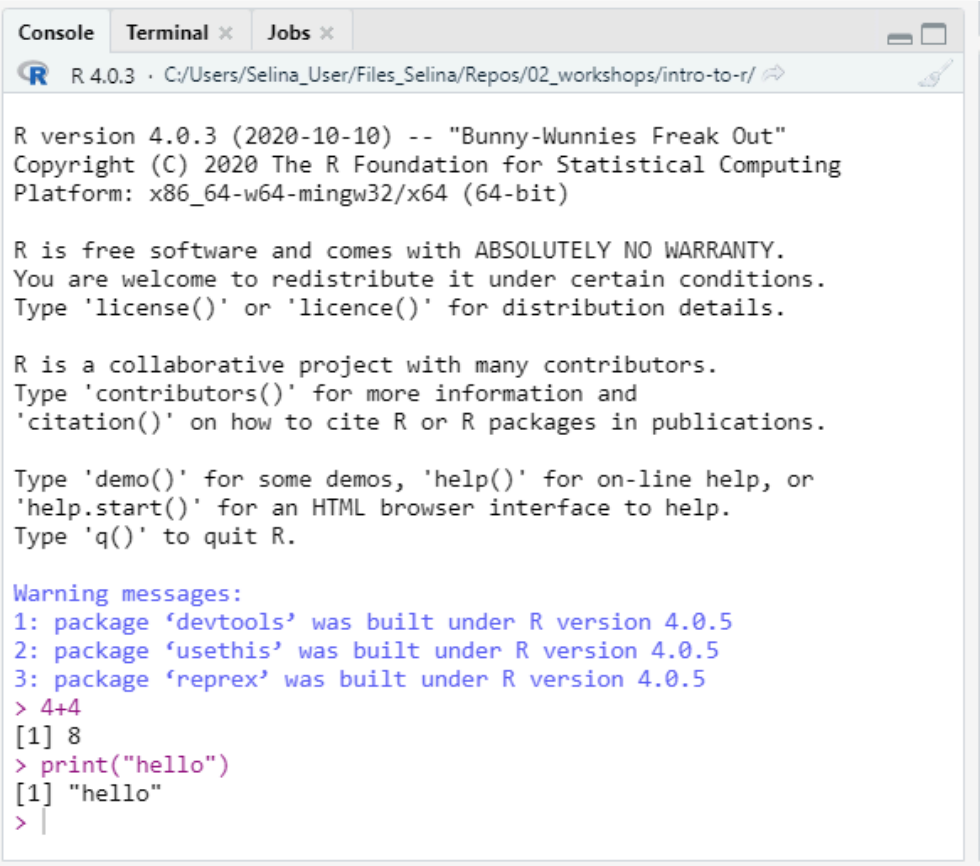
# A quick tour around RStudio



# Console pane

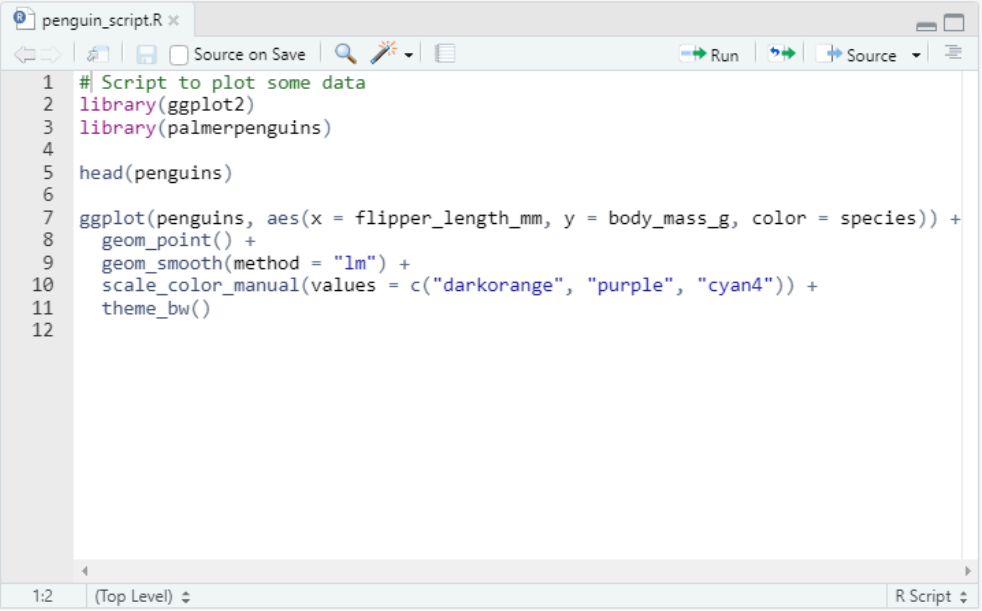
- Execute R code
- Output from R code in scripts is printed there
- Type a command into the console and execute with `Enter/Return`

> **Tip**
>
> Use arrow keys to bring back last commands



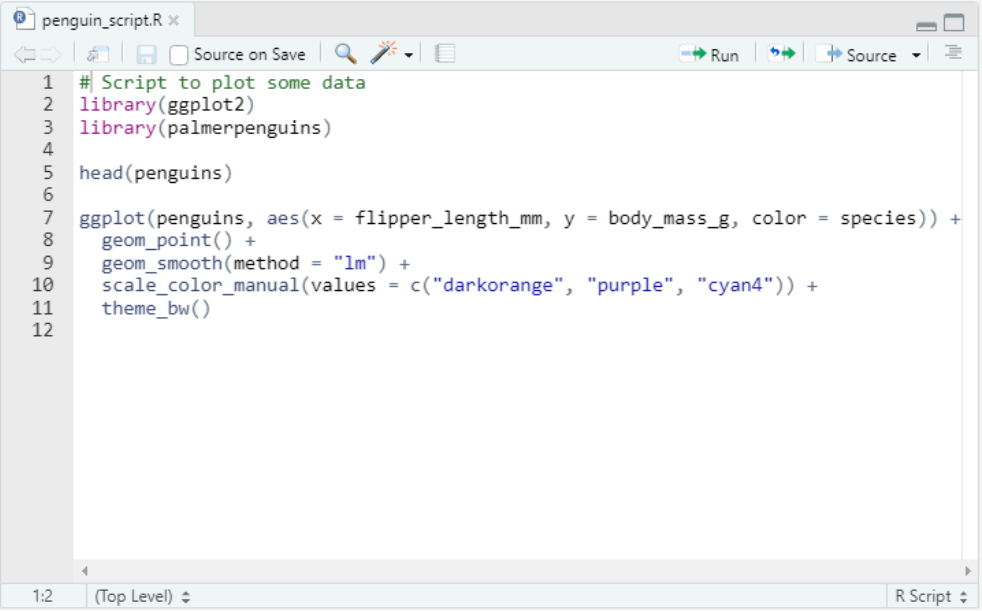
# Script pane

- Write scripts with R code
  - Scripts are text files with R commands (file ending `.R`)
  - Use scripts to save commands for reuse



# Script pane

- Create a new R script:
  **File -> New File -> R Script**
- Save an R script:
  **File->Save (Ctrl/Cmd + S)**
- Run code line by line with **Run button (Ctrl+Enter/Cmd+Return)**
- You can open multiple scripts

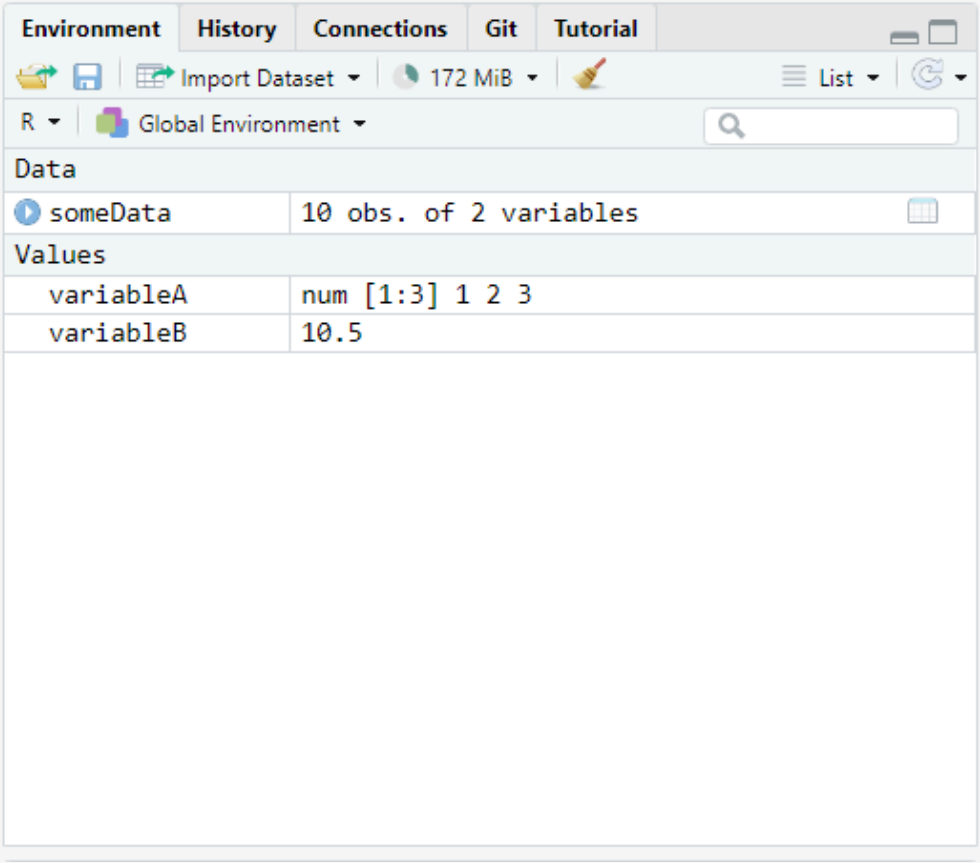



**Script vs Console**

Use **scripts** for all your analysis and for commands that you want to save.
Use **console** for temporary commands, e.g. to test something.

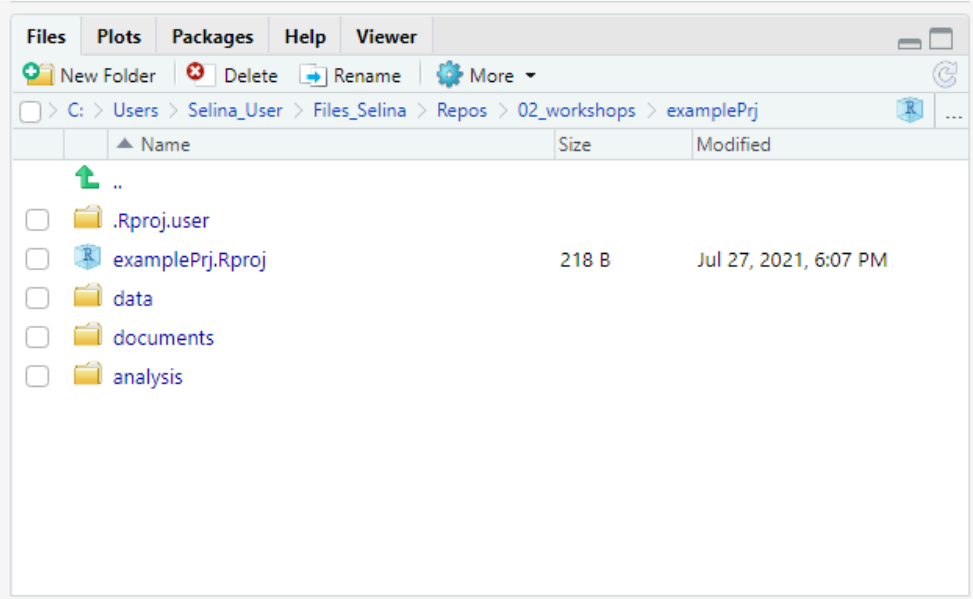
# Environment pane

- Shows objects currently present in the R session
- Is empty if you start R



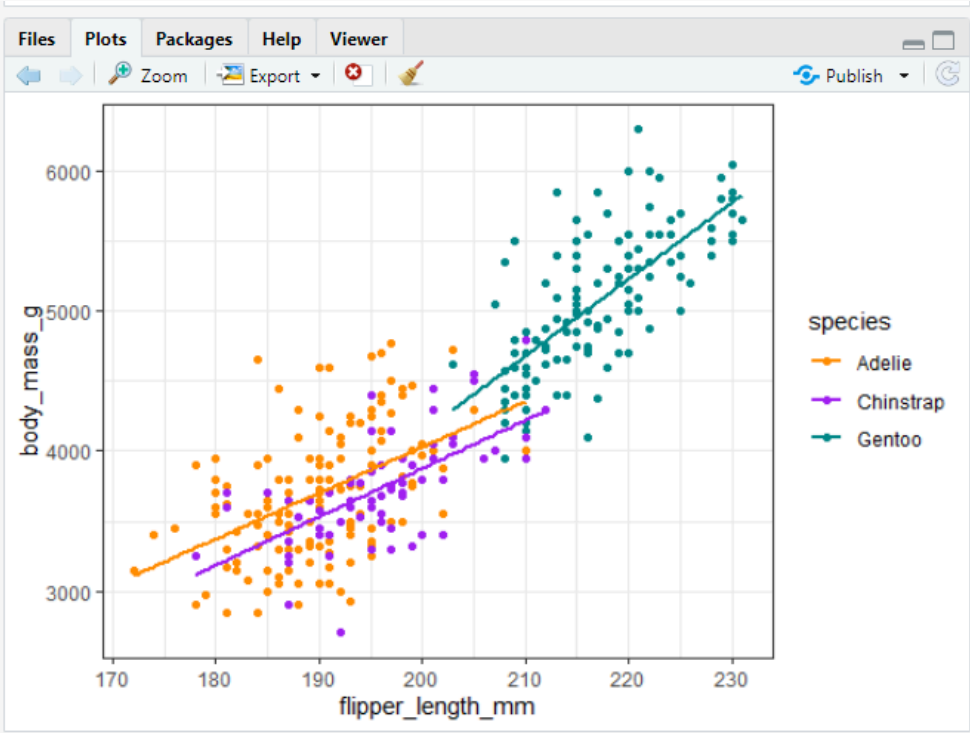
# Files pane

- Similar to Explorer/Finder
- Browse project structure and files
  - Find and open files
  - Create new folders
  - Delete and rename files
  - …



# Plot pane

- Plots that are created with R will be shown here



# A tip before we get started

Learn the most important keyboard shortcuts of RStudio.

Find all shortcuts under **Tools -> Keyboard Shortcuts Help**

- Save active file: Ctrl/Cmd + S
- Run current line: Ctrl/Cmd + Enter
- Create new R Script: Ctrl/Cmd + N
- Undo: Ctrl/Cmd + Z
- Redo: Ctrl/Cmd + Y
- Copy/Paste: Ctrl/Cmd + C/V

# RStudio projects

Idea: One directory (folder) with all files relevant for project (data, R scripts, figures, ...)

- An **RStudio project** is just a normal folder with an **.Rproj** file
- Helps R to find your project files

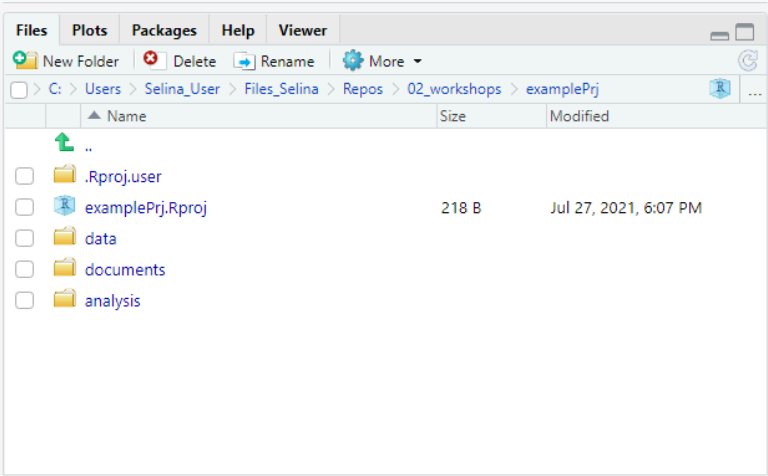
```
MyProject
|
|- data
|
|- R
|    |
|    |- clean_data.R
|    |
|    |- statistics.R
|
|- figures
|
|- MyProject.RProj
```

Example RStudio project structure

# Create an RStudio project

Create a project from scratch:

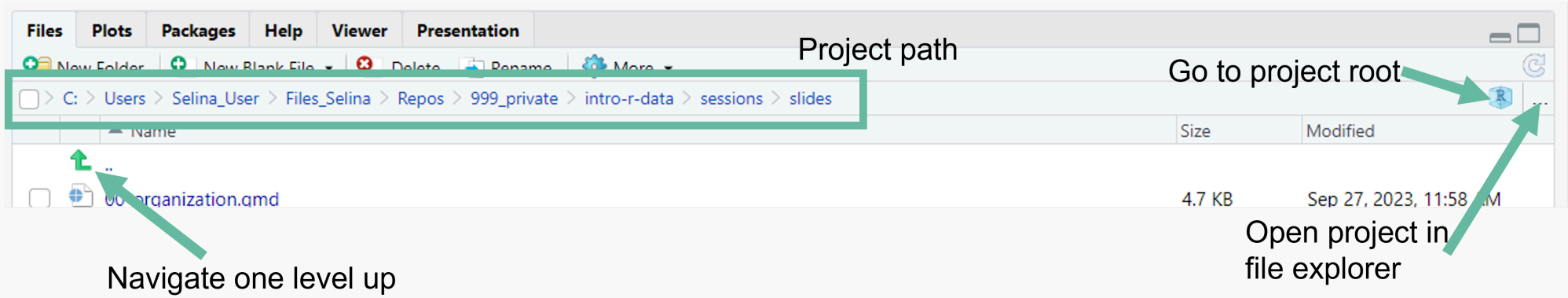
1. **File -> New Project -> New Directory -> New Project**
2. Enter a directory name (this will be the name of your project)
3. Choose the Directory where the project should be initiated
4. **Create Project**




Example RStudio project structure in the Files pane

RStudio will now create and open the project for you.

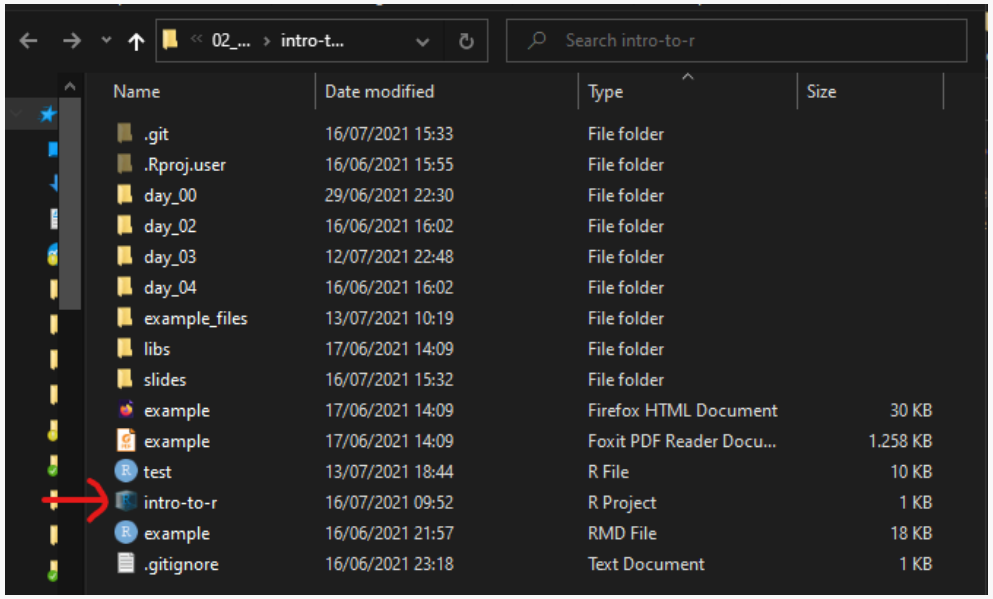
# Navigate an RStudio project



# Open an RStudio project

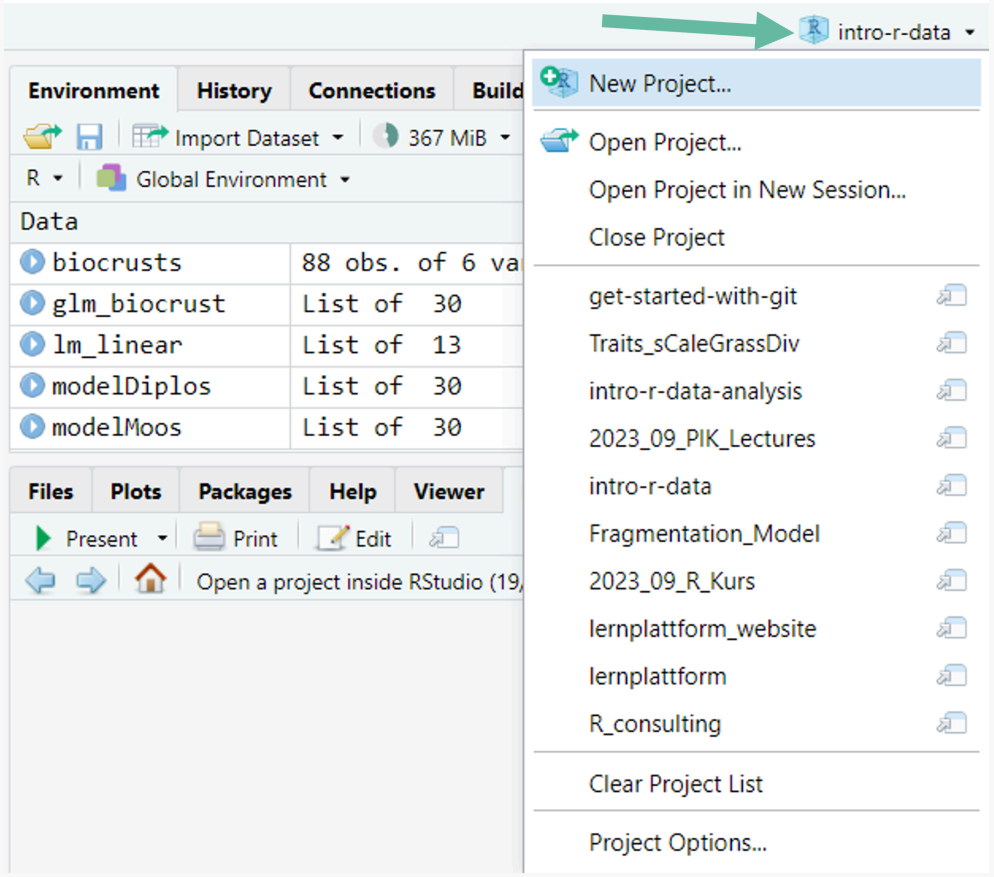
## From file explorer/finder:

Double click on the **.Rproj** file




## From inside RStudio:

Click the project symbol on the top right



# Now you

Task 1 (15 min)

Set up your own RStudio project for this workshop

Find the task description here